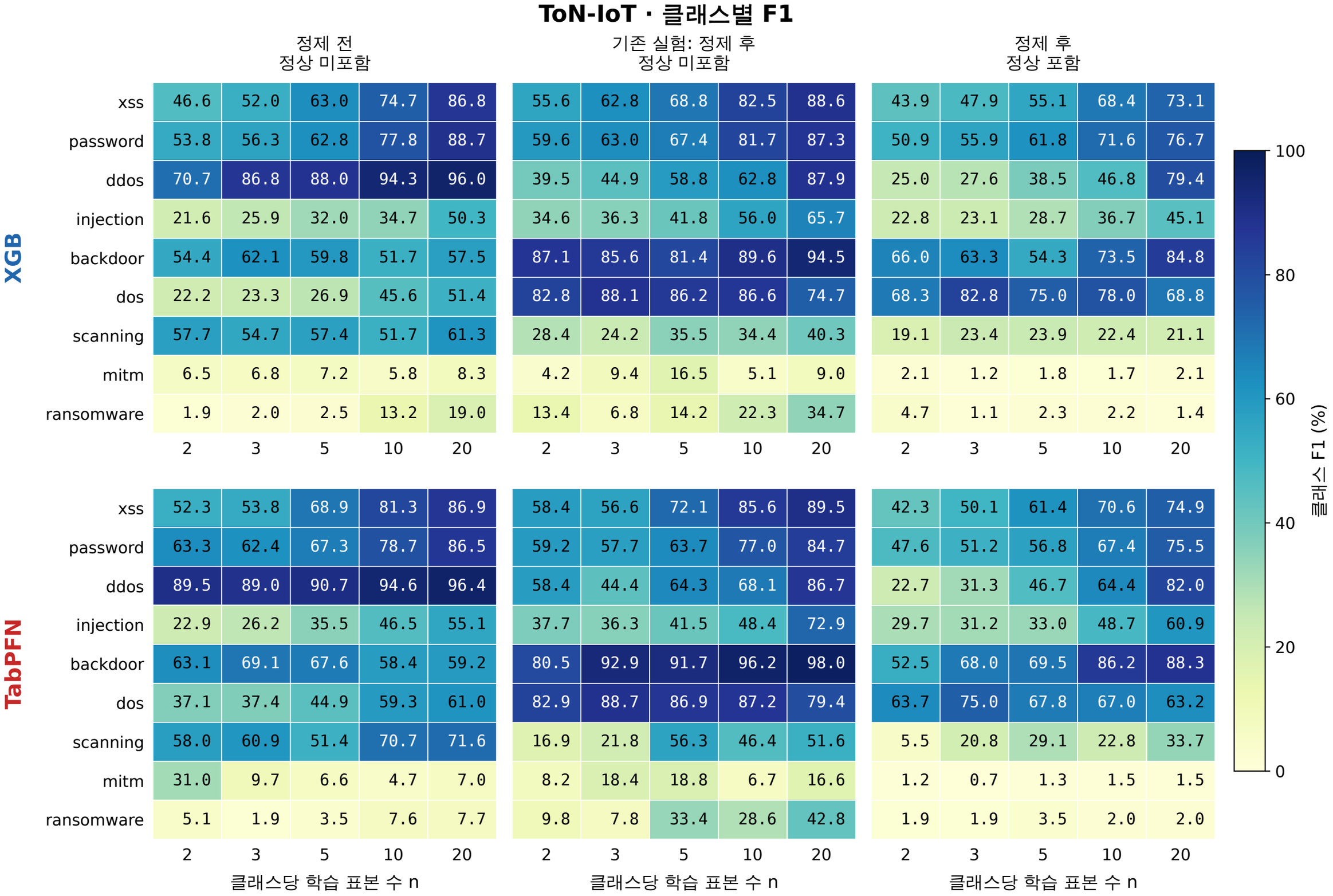
ToN-IoT · 클래스별 F1
정제 전
정상 미포함
기존 실험: 정제 후
정상 미포함
정제 후
정상 포함
XGB
TabPFN
xss 46.6 52.0 63.0 74.7 86.8 55.6 62.8 68.8 82.5 88.6 43.9 47.9 55.1 68.4 73.1
password 53.8 56.3 62.8 77.8 88.7 59.6 63.0 67.4 81.7 87.3 50.9 55.9 61.8 71.6 76.7
ddos 70.7 86.8 88.0 94.3 96.0 39.5 44.9 58.8 62.8 87.9 25.0 27.6 38.5 46.8 79.4
injection 21.6 25.9 32.0 34.7 50.3 34.6 36.3 41.8 56.0 65.7 22.8 23.1 28.7 36.7 45.1
backdoor 54.4 62.1 59.8 51.7 57.5 87.1 85.6 81.4 89.6 94.5 66.0 63.3 54.3 73.5 84.8
dos 22.2 23.3 26.9 45.6 51.4 82.8 88.1 86.2 86.6 74.7 68.3 82.8 75.0 78.0 68.8
scanning 57.7 54.7 57.4 51.7 61.3 28.4 24.2 35.5 34.4 40.3 19.1 23.4 23.9 22.4 21.1
mitm 6.5 6.8 7.2 5.8 8.3 4.2 9.4 16.5 5.1 9.0 2.1 1.2 1.8 1.7 2.1
ransomware 1.9 2.0 2.5 13.2 19.0 13.4 6.8 14.2 22.3 34.7 4.7 1.1 2.3 2.2 1.4
2 3 5 10 20 2 3 5 10 20 2 3 5 10 20
xss 52.3 53.8 68.9 81.3 86.9 58.4 56.6 72.1 85.6 89.5 42.3 50.1 61.4 70.6 74.9
password 63.3 62.4 67.3 78.7 86.5 59.2 57.7 63.7 77.0 84.7 47.6 51.2 56.8 67.4 75.5
ddos 89.5 89.0 90.7 94.6 96.4 58.4 44.4 64.3 68.1 86.7 22.7 31.3 46.7 64.4 82.0
injection 22.9 26.2 35.5 46.5 55.1 37.7 36.3 41.5 48.4 72.9 29.7 31.2 33.0 48.7 60.9
backdoor 63.1 69.1 67.6 58.4 59.2 80.5 92.9 91.7 96.2 98.0 52.5 68.0 69.5 86.2 88.3
dos 37.1 37.4 44.9 59.3 61.0 82.9 88.7 86.9 87.2 79.4 63.7 75.0 67.8 67.0 63.2
scanning 58.0 60.9 51.4 70.7 71.6 16.9 21.8 56.3 46.4 51.6 5.5 20.8 29.1 22.8 33.7
mitm 31.0 9.7 6.6 4.7 7.0 8.2 18.4 18.8 6.7 16.6 1.2 0.7 1.3 1.5 1.5
ransomware 5.1 1.9 3.5 7.6 7.7 9.8 7.8 33.4 28.6 42.8 1.9 1.9 3.5 2.0 2.0
2 3 5 10 20 2 3 5 10 20 2 3 5 10 20
클래스당 학습 표본 수 n
클래스당 학습 표본 수 n
클래스당 학습 표본 수 n
100
80
60
40
20
0
클래스 F1 (%)